MIL-P-45637B(MU)
11 January 1965
SUPERSEDING
MIL-P-45637A(Ord)
11 April 1960

MILITARY SPECIFICATION

PERISCOPES, TANK: M19 and M24

1. SCOPE

1.1 This specification covers two types of infrared binocular instruments known as Periscope, Tank: M19 and Periscope, Tank: M24. Both are similar except that Periscope M24 has an extended body.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 6650

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DRAWINGS

U.S. Army Munitions Command

F7652971	Periscope, Tank: M19
F8293676	Periscope, Tank: M24
IEL 7652971	List of Inspection Documents for Periscope, Tank: M19
IEL 8293676	List of Inspection Documents for Periscope, Tank: M24

PACKAGING DATA SHEETS

MIL-P-14232/P7652971	Packaging of Periscope, Tank: M19
MIL-P-14232/P8293676	Packaging of Periscope, Tank: M24

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The periscope shall be manufactured in accordance with Drawing F7652971 for Periscope, Tank: M19, Drawing F8293676 for Periscope, Tank: M24, and drawings pertaining thereto.

3.3 General specifications.- The periscope shall meet the following requirements of Specifications MIL-O-13830 and MIL-F-13926.

3.3.1 MIL-O-13830.

(a) Cleanliness
(b) Fixed eyepiece focus

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3.3.2 MIL-P-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Electrical and electronic assemblies
(e) Part identification and marking
(f) Workmanship

3.4 Environmental.

3.4.1 Storage temperatures.- The periscope shall show no evidence of physical failure and shall meet all performance requirements of this specification at a standard ambient temperature of 60° to 90° Fahrenheit (F) after having been exposed to and thermally stabilized at ambient temperatures of -80° and +160°F.

3.4.2 Vibration.- Prior to vibration, the periscope shall meet the requirements of 3.6.4 and these values shall be recorded. The periscope shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ±15 seconds.

3.4.2.1 Post-vibration.- Subsequent to vibration, there shall be no evidence of physical damage. The optical elements in the periscope shall show no more than 5 particles of dirt one mil maximum apparent diameter in the central 10 degrees of the apparent field, and no more than 10 particles of dirt 1.5 mils maximum apparent diameter in the remainder of the field. Dirt on surfaces more than two diopters out of the focal plane shall not exceed the size and number of digs permitted for that surface. The periscope shall again meet the requirements of 3.6.4. The divergence reading shall be within 10 minutes and the dipvergence reading shall be within 5 minutes of the values recorded prior to vibration.

3.4.3 Sealing.- The periscope shall be externally pressurized to 5 pounds per square inch gage (psig) and shall show no evidence of leakage after a period of 5 minutes.

3.4.4 Rain.- There shall be no evidence of water within the cavities at the top of the body just below the head of the periscope after being exposed to Procedure I of the rain test specified in Specification MIL-P-13926.

3.5 Electrical.

3.5.1 Continuity.- The periscope shall demonstrate continuity in accordance with the wiring diagram.

3.5.2 Dielectric strength.- There shall be no evidence of insulation breakdown when a potential of 20,000 volts dc ±500 volts is applied between the inner conductor and ground of the power receptacle for a period of not less than one minute with the image converter infrared sensitive tubes removed, the ground leads removed, and the terminal housings in place.

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3.5.3 Ground potential.- The external parts of the periscope shall be at ground potential at all times.

3.6 Performance.- A potential of 16,000 volts dc ±100 volts shall be applied to the periscope to determine compliance with the following requirements.

3.6.1 Electrostatic focus.- The electrostatic focus potentiometers shall have sufficient movement to go through the sharpest or best focus position. The output voltage of the focus potentiometers shall cover a range from 2120 to 2600 volts.

3.6.2 Eyepiece focus.- The eyepieces shall be set between -0.75 and -1.0 diopter. The electrostatic focus shall be set so that the image on the fluorescent screen will be in the sharpest focus at the center of the field. No further electrostatic adjustment of focus shall be permitted for the requirements specified in 3.6.3 to 3.6.6 inclusive.

3.6.3 Resolution.- The resolving power, within 2 degrees of the optical axis for a parallel bar test target at 18 to 20 yards, shall be 4 minutes of arc or less.

3.6.4 Collimation.- With the rays from an object point 18 to 20 yards distant projected into the two objectives simultaneously, the conjugate rays emerging from the two eyepieces shall be divergent in a horizontal plane by 3 degrees 11 minutes ±40 minutes measured at the eyepoint and at an interpupillary distance of 65 millimeters. Dipvergence of the target shall not exceed 15 minutes of arc.

3.6.4.1 Collimation change due to electron image shift.- The periscope shall be subjected to ten cycles of "on" and "off" at the rated voltage; the voltage being applied for 15 seconds and removed for 10 seconds. Collimation of the periscope at the expiration of each "on" period shall not exceed the requirements of 3.6.4.

3.6.5 Paraxial magnification.- The paraxial magnification of the two optical systems shall not differ by more than six percent. The paraxial magnification shall be measured at the same points in the field in each eyepiece, i.e., at the edges of the 0.161 inch diameter field of the eyepiece.

3.6.6 Image tilt.- The image of a plumb line formed by each of the two optical systems shall be vertical within two degrees of arc and shall be parallel to each other within one degree.

3.6.7 Image steadiness.- There shall be no visible flicker, no movement of the image, nor audible sparking when the image is observed for one minute after the expiration of 15 seconds tube stabilization time.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The quality assurance provisions of this specification and of other documents referenced in this specification and in the contract shall constitute the minimum amount of inspection performed by the contractor prior to submission of the product for Government acceptance, and will constitute the maximum severity to be used by the Government for acceptance or rejection of any submitted lot. Inspection of components and sub-assemblies not covered by specifications or Supplementary Quality Assurance Provisions (SQAP) shall be in accordance with 4.2.2. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" specified in Standard MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-P-13926 and Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents, IEL-7652971 and IEL-8293676 as applicable. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105 and the inspection level and sampling plans specified in Table I titled "Classification of Defects". Examinations and tests for packaging, packing, and marking shall be in accordance with Specification MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

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TABLE I - CLASSIFICATION OF DEFECTS

Use General Inspection Level II of Table I with
Sampling Plan Table II-A of Standard MIL-STD-105

		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 1.0 percent defective			
101.	Collimation, pre-vibration	3.4.2, 3.6.4	4.5.2, 4.5.10.4
102.	Vibration	3.4.2	4.5.3
103.	Cleanliness	3.4.2.1	4.5.3.2
104.	Collimation, post vibration	3.4.2.1, 3.6.4	4.5.3.3, 4.5.10.4
105.	Sealing	3.4.3	4.5.4
106.	Electrostatic focus	3.6.1	4.5.10.1
107.	Eyepiece focus	3.6.2	4.5.10.2
108.	Resolution	3.6.3	4.5.10.3
109.	Collimation change due to electron image shift	3.6.4.1	4.5.10.5
110.	Paraxial magnification	3.6.5	4.5.10.6
111.	Image tilt	3.6.6	4.5.10.7
112.	Image steadiness	3.6.7	4.5.10.8
MINOR: None defined			

NOTE: The tests of Table I shall be conducted at a standard ambient temperature (plus 60 degrees to plus 90 degrees Fahrenheit).

4.2.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" specified in Standard MIL-STD-105.

4.3 Control tests.- One periscope of each model being procured shall be selected at random from each 100 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Fabrication	3.2	Visual - per drawing
General specifications	3.3	Visual - MIL-O-13830, MIL-F-13926

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4.3.1 Control tests for electrical requirements.- Electrical examinations and tests shall be conducted prior to final sealing of the periscope. Three periscopes of each model being procured shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. The control samples shall be subjected to the following examinations and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Continuity	3.5.1	4.5.6
Dielectric strength	3.5.2	4.5.7
Ground potential	3.5.3	4.5.8
Potentiometer output voltage	3.6.1	4.5.9

4.3.2 Control tests for environmental requirements.- Three periscopes of each model being procured shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Storage temperature	3.4.1	4.5.1
Rain	3.4.4	4.5.5

4.3.3 Control test failure.- Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall maintain all required measuring and testing equipment in accordance with the applicable requirements specified in Specification MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in Specification MIL-I-45607.

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4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the requirements of 4.4 and the List of Inspection Documents, IEL-7652971 for Periscope: M19 and IEL-8293676 for Periscope: M24, as applicable.

4.4.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" requirements of Specification MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5 Test methods and procedures.

4.5.1 Storage temperature.- Test shall be in accordance with the "Test Facilities" requirements of Specification MIL-F-13926 and the conditions of 3.4.1. Upon return to standard ambient temperature, the periscope shall be subjected to a visual and tactile examination and shall show no evidence of electronic, mechanical, or optical failure. The periscope shall then be subjected to the tests specified herein, and shall conform to the requirements specified in 3.4.1.

4.5.2 Collimation, pre-vibration.- The periscope shall be inspected for collimation in accordance with 4.5.10.4 prior to vibration. Collimation of the periscope shall conform to the requirements of 3.6.4. Record the pre-vibration collimation values obtained for evaluation against the post vibration collimation values, to determine conformance to the requirements of 3.4.2.1.

4.5.3 Vibration.- Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.4.2 shall be utilized for the vibration test. The periscope shall be vibrated in the vertical plane. Unless otherwise specified, the test equipment shall be in conformance with the "Test Facilities" requirements of Specification MIL-F-13926.

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4.5.3.1 Physical damage, post vibration.- Subsequent to vibration the periscope shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4.2 prior to being subjected to the post vibration collimation test and the remaining tests specified herein. A visual examination shall be made for missing or damaged parts prior to being subjected to the remaining performance tests. Minimum visual examination shall include elevation lock assembly, caps, chains, nuts, screws, head assembly, and eyepiece assemblies. A tactile examination should also be conducted to determine looseness or improper function of the above items and supporting or assembled members, including the head latching assembly. An audible examination shall be made by holding the periscope close to the ear and vigorously shaking the periscope with both hands while listening for loose internal optical, electronic, and mechanical parts (lenses, prisms, cells, electron tubes, retainers, cables, bonded elements, etc.). Any one of the possible defects defined above shall be cause to consider a periscope defective unless separate defects-per-hundred type sampling plans are established and approved by the Government as part of the suppliers inspection system. Periscopes which have passed the above examinations shall be subjected to the test of 4.5.3.2.

4.5.3.2 Cleanliness, post vibration.- Position the periscope and energise the image converter tubes as specified in 4.5.10. Visually inspect the optical surfaces in each system for cleanliness. "Hot" and "cold" spots on the fluorescent screen of the tubes will appear as light or dark spots and are not to be considered as dirt. To differentiate between "hot" and "cold" spots and dirt, decrease and increase the output of the power supply while sighting through the periscope. This will cause the irregularities in the image converter tube to change from light to dark spots or from dark to light spots, while dirt will always remain dark. Exercise caution during variation of voltage so as not to exceed 16,500 volts input to the periscope. Dirt in excess of the tolerance specified in 3.4.2.1 is considered faulty workmanship and is cause for rejection. Periscopes which have passed the above examinations shall be subjected to the test of 4.5.3.3.

4.5.3.3 Collimation, post vibration.- Retest the periscope for collimation in accordance with 4.5.10.4. Collimation of the periscope shall conform to the requirements of 3.6.4. Record the post vibration collimation values. Any change in collimation, which is evidenced as a difference between the recorded pre-vibration and post vibration values, shall not exceed the dipvergence and divergence tolerance specified in 3.4.2.1.

4.5.4 Sealing.- Special testing equipment that conforms to Drawing F7680673, Fixture, Pressure Testing; together with Drawing F7297923, Adapter for Periscope M24; or Drawing F7560062, Adapter for Periscope M19; shall be utilised for the applicable periscope under test. The test procedure for performing the sealing test shall be as detailed in the operating instructions on Drawing F7680673. The periscope shall conform to the sealing requirements of 3.4.3. The sealing test shall be performed subsequent to the final sealing of the periscope.

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4.5.5 Rain test.- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926. While exposed to the simulated rainfall the periscope shall be secured in its using position, and the high voltage connector shall be sealed against moisture penetration. Immediately following the specified exposure the head assembly shall be removed, and the cavities in the top of the body shall be inspected for moisture penetration. The periscope shall conform to the requirements of 3.4.4. To facilitate inspection, the external surfaces of the periscope may be dried by compressed air or blotting with water-absorbing material prior to removal of the head assembly. Upon completion of this test the periscope shall be thoroughly dried.

4.5.6 Continuity.- Standard measuring equipment shall be utilized for this test. The electrical circuits of the periscope shall conform to the requirements of 3.5.1. Perform this test prior to final sealing of the periscope.

4.5.7 Dielectric strength.- Conduct this test in accordance with Method 301 of Standard MIL-STD-202, with the image tubes removed from the periscope and the terminal board ground wires disconnected from the body. A high-voltage power source that conforms to Drawing F7561204 shall be utilized to supply the voltage for this test. During test voltage application a visual and audible examination shall be conducted on the periscope circuits for evidence of grounding. Subsequent to voltage application the periscope circuits shall be subjected to a visual examination for evidence of insulation breakdown and damage due to corona. The periscope shall conform to the requirements of 3.5.2. The dielectric strength test shall be performed prior to the final sealing of the periscope.

4.5.8 Ground potential.- Standard measuring equipment shall be utilized for this test. The focusing adjustment shaft of the potentiometers shall be included in this test. The periscope shall conform to the requirements of 3.5.3.

4.5.9 Potentiometer output voltage, electrostatic focus.- Perform this test subsequent to installation of the image tubes and prior to final sealing of the periscope. Standard measuring equipment shall be utilized for measuring the output voltage of each focusing potentiometer. A high-voltage power source that conforms to Drawing F7561204 shall be utilized to energize the periscope. The voltage applied to the periscope shall be as specified in 3.6. The output voltage of each focusing potentiometer shall conform to the requirements of 3.6.1 when the potentiometer shaft is rotated from the clockwise stop to the counterclockwise stop.

4.5.10 Performance.- Special testing equipment that conforms to Drawing F7560189 shall be utilized in testing the performance characteristics of the periscope. The voltage applied to energize the periscope shall conform to the requirements of 3.6, and shall be supplied by a high-voltage power source that conforms to Drawing F7561204. The high-voltage power source shall have a 2 minute warm-up period prior to applying the operating voltage to the periscope. All performance tests shall be conducted subsequent to final sealing of the periscope.

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4.5.10.1 Electrostatic focus.- Direct a line of sight through the left eyepiece of the periscope to the resolution target image in the upper collimators of the testing equipment while simultaneously rotating the left focus potentiometer shaft from its clockwise stop to its counterclockwise stop. At one position during this shaft rotation, before contact with the counterclockwise stop, the target image shall appear to be in sharpest or best focus. Repeat this test on the right focus potentiometer. The electrostatic focus potentiometers, left and right, shall conform to the requirements of 3.6.1 for best focus position.

4.5.10.2 Eyepiece focus.- A calibrated dioptometer having a magnification of at least three power (3X) shall be utilized for measuring the eyepiece focus of each optical system in the periscope. Set the dioptometer diopter scale to zero, and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Position the dioptometer at each eyepiece of the periscope, in turn, and obtain best focus of the phospor grain on the screen of the image tube by adjustment of the dioptometer diopter scale. The phosphor grain on the screen of the image tube shall be in best focus, in each optical system, within the diopter tolerance specified in 3.6.2, as measured on the dioptometer diopter scale. While the dioptometer is positioned at each eyepiece, and set to the measured focus of the eyepiece, adjust the shaft of the focus potentiometer and obtain the sharpest focus of the resolution target image in the upper collimators of the testing equipment. Record the measured focus of each eyepiece. The restrictions on electrostatic focusing specified in 3.6.2 shall be observed.

4.5.10.3 Resolution.- Perform this test in sequence to the test for eyepiece focus. Focus the dioptometer by means of the diopter scale adjustment, and obtain correct line count of the resolution target image in all four meridians. Resolution of the target image, in all four meridians, shall be obtained within 1/8 diopter of the measured eyepiece focus value obtained for the eyepiece in the periscope. The resolution of each optical system in the periscope shall be tested.

4.5.10.4 Collimation.- Perform this test with the lower collimators in position at the periscope eyepieces.

4.5.10.4.1 Divergence.- Divergence of 3 degrees 11 minutes is "built into" the lower collimator assembly of the testing equipment when it is calibrated in accordance with the set-up instructions on Drawing F7560189. The amount of divergence greater than, or less than, 3 degrees 11 minutes shall be measured by means of the horizontal scale divisions on the center of the reticle in the lower collimator, with respect to the vertical reticle line in the upper collimator. The scale divisions represent 10 minutes of arc, for a total of 40 minutes of arc on each side of the vertical reticle line. Adjust the lower gimbal mounting to obtain coincidence of the centers of the lower and upper right collimator reticle patterns. Read the amount of divergence greater than, or less than, the established 3 degrees 11 minutes on the reticle scale in the lower left collimator. The conjugate rays emerging from the two eyepieces shall be divergent in a horizontal plane within the tolerance specified in 3.6.4. Record divergence value.

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4.5.10.4.2 Dipvergence.- Dipvergence of zero minutes is "built into" the lower collimator assembly of the testing equipment when it is calibrated in accordance with the set-up instructions on Drawing F7560189. Dipvergence shall be measured by means of the center scale divisions on the vertical reticle line in the lower collimator, with respect to the horizontal reticle line in the upper collimator. The scale divisions represent 5 minutes of arc, for a total of 15 minutes of arc on each side of the horizontal reticle line. Adjust the lower gimbal mounting as in 4.5.10.4.1 and read the amount of dipvergence on the reticle scale in the lower left collimator. The conjugate rays emerging from the two eyepieces shall not exceed the dipvergence tolerance specified in 3.6.4. Record dipvergence value.

4.5.10.5 Collimation change due to electron image shift.- This test shall require a suitable "off-on" cycling switch to be inserted in the high voltage line between the high voltage power supply and the periscope. The power supply unit shall not be cycled for this test. Adjust the lower gimbal mounting as in 4.5.10.4.1. Subject the periscope to the "on-off" voltage cycling specified in 3.6.4.1, while directing a line of sight through the lower collimators. At the expiration of each "on" period the collimation of the periscope shall conform to the requirements of 3.6.4.1, as measured on the lower collimator reticles with reference to the upper collimator reticles.

4.5.10.6 Paraxial magnification.- The bow tie reticle pattern in the upper collimators, in conjunction with the upper vertical reticle scale in the lower collimators, shall be utilized for measuring magnification of the periscope. Adjust the lower gimbal mounting as in 4.5.10.4.1. Record the point of intersection of the bow tie reticle pattern image with the vertical reticle scale image as the magnification value for the periscope's right optical system. Aline the lower left collimator with the upper collimator, as above, and obtain the magnification value of the periscope's left optical system. Each division on the vertical reticle scale represents 2 percent. Any difference in magnification between the left and right optical systems shall not exceed the tolerance specified in 3.6.5.

4.5.10.7 Image tilt.- The horizontal scale at the bottom of the reticle in the lower collimators, in conjunction with the vertical reticle line in the upper collimators, shall be utilized for measuring image tilt in each optical system of the periscope. Each division on the horizontal reticle scale represents 30 minutes of arc. Adjust the lower gimbal mounting as in 4.5.10.4.1. Record the point of intersection of the vertical reticle line image with the horizontal scale image as the image tilt value for the right optical system. Aline the lower left collimator with the upper collimator, as above, and record the image tilt value for the left optical system. Any image tilt in the optical systems of the periscope shall not exceed the tolerances specified in 3.6.6.

4.5.10.8 Image steadiness.- This test shall consist of a simultaneous visual examination of the image screens and an audio examination of the periscope when it is energized for the specified time. The periscope shall conform to the requirements of 3.6.7.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheets MIL-P-14232/P7652971, Packaging of Periscope, Tank: M19, and MIL-P-14232/P8293676, Packaging of Periscope, Tank: M24. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The infrared periscopes are used to facilitate night driving of tanks, self-propelled vehicles, and armored personnel carriers.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (see 5.1).

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 6650-A147Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO

QUANTITY OF ITEM PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE

PLATE NO 15419

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING OFFICER
FRANKFORD ARSENAL
ATTN: SMUFA-N1100
PHILA., PA. 19137

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